AA–TV wiring v7, PLAN A: front driven wheels. Deutsch DT connectors only (wet zone); cables epoxy–potted into dryboxes; 14 AWG / 20 AWG.

BUOY (surface, dry)

BT2
USB–C power bank 5 V / 3 A
5V_PI
GND_PI

J1

GPIO12/PWM0		5V	2 — 5V_PI
GPIO13/PWM1		3V3	1
GPIO14/UART_TXD		ID_SD_I2C0/GPIO00	27
GPIO15/UART_RXD		ID_SC_I2C0/GPIO01	28
GPIO16/SPI1_CE2		SDA_I2C1/GPIO02	3
GPIO17/SPI1_CE1		SCL_I2C1/GPIO03	5
GPIO18/SPI1_CE0/PCM_CLK/PWM0			
GPIO19/SPI1_MISO/PCM_FS		GPCLK0/GPIO04	7
GPIO20/SPI1_MOSI/PCM_DIN/PWM1		GPCLK1/GPIO05	29
GPIO21/SPI1_SCLK/PCM_DOUT		GPCLK2/GPIO06	31
GPIO22/SDIO_CLK		CE1_SPI0/GPIO07	26
GPIO23/SDIO_CMD		CE0_SPI0/GPIO08	24
GPIO24/SDIO_DAT0		MISO_SPI0/GPIO09	21
GPIO25/SDIO_DAT1		MOSI_SPI0/GPIO10	19
GPIO26/SDIO_DAT2		SCLK_SPI0/GPIO11	23
GPIO27/SDIO_DAT3			
	GND		6 — GND_PI

Raspberry Pi 4 (buoy)

Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power–cycling the Pi USB ports (uhubctl).

TETHER

W1
USB–A — micro–B
5V_PI — + USB data — VBUS_PICO
GND_PI — GND

USB–A to micro–B, ~3 m

Vinyl air tube from the buoy; USB cable runs inside it. USB plugs are dry (buoy, Pico).

PEN1
A1: air–tube fitting (USB inside)

MAIN DRYBOX (upper, on hull)

BOARD A: Pico solder–board

GP12	R1 220	M4_PWM
GP13	R2 220	M5_PWM
GP14	R3 220	M6_PWM
GP15	R4 220	M7_PWM
GP16	R5 220	THR_PWM
GP19	R6 220	PUMP_PWM
GP17	R7 220	DRV_L_PWM
GP18	R8 220	DRV_R_PWM

Raspberry Pi Pico A1

VBUS_PICO	40		
+3V3	36	3V3	
	39	VSYS	
	30	RUN	
	37	3V3_EN	
	1	GPIO0	
	2	GPIO1	
	4	GPIO2	
	5	GPIO3	
	6	GPIO4	
	7	GPIO5	
	9	GPIO6	
	10	GPIO7	
	11	GPIO8	
	12	GPIO9	
	14	GPIO10	
	15	GPIO11	
GP12	16	GPIO12	
GP13	17	GPIO13	
GP14	19	GPIO14	
GP15	20	GPIO15	
	21	GPIO16	GP16
	22	GPIO17	GP17
	24	GPIO18	GP18
	25	GPIO19	GP19
	26	GPIO20	
	27	GPIO21	
	29	GPIO22	
	35	ADC_VREF	
	31	GPIO26_ADC0	LEAK_MAIN
	32	GPIO27_ADC1	LEAK_REAR
	34	GPIO28_ADC2	
	33	AGND	GND
	3	GND	

220 ohm in series on every PWM line that leaves the drybox.

TA2
1 +3V3
2 GND
3 LEAK_MAIN

Main leak sensor

TA1

M4_PWM	1
M5_PWM	2
M6_PWM	3
M7_PWM	4
THR_PWM	5
DRV_L_PWM	6
DRV_R_PWM	7
PUMP_PWM	8
LEAK_REAR	9
+3V3	10
GND	11
GND	12

All Board A terminals: 5.08 mm pitch (0.2 in, fits the breadboard grid).

H1 out (20 AWG)

PEN2
A2: H1 cable, epoxy–potted

REAR DRYBOX (inside open rear endcap)

BOARD B: servo / signal board

TB1

M4_PWM	1
M5_PWM	2
M6_PWM	3
M7_PWM	4
THR_PWM	5
DRV_L_PWM	6
DRV_R_PWM	7
PUMP_PWM	8
LEAK_REAR	9
+3V3	10
GND	11
GND	12

H1 in (20 AWG)

TB6
1 SERVO_7V5
2 GND
7.5 V in (14 AWG)

C1
SERVO_7V5 — GND
1000uF 16V

Board B:
– 7.5 V + GND rails: strip of 14 AWG soldered along strips
– No 12 V on the breadboard
– 5.08 mm terminals only

TB2

M4_PWM	1
SERVO_7V5	2
GND	3
M5_PWM	4
SERVO_7V5	5
GND	6
M6_PWM	7
SERVO_7V5	8
GND	9
M7_PWM	10
SERVO_7V5	11
GND	12

Servo cable (20 AWG)

TB3
THR_PWM 1
GND 2
Thruster ESC signal

TB4
PUMP_PWM 1
GND 2
Pump ESC signal

TB5
+3V3 1
GND 2
LEAK_REAR 3
Rear leak sensor

TB7
DRV_L_PWM 1
GND 2
DRV_R_PWM 3
GND 4
Wheel ESC signals

Off–board, in rear drybox

BB1
VBAT — 1 2 3 4 5 6 7 8
Bus bars (screw studs, ring terminals):
+ in: D1 pins 1+2 (two 14 AWG)
+ out: buck, pump ESC, D2 thruster, D5 wheel L, D5 wheel R
GND bar mirrors the + bar.
+12 V bus bar (>=30 A)

BB2
GND — 1 2 3 4 5 6 7 8
GND bus bar (>=30 A)

U1 BUCK
VBAT 1 — VIN+ — VOUT+ — 3 SERVO_7V5
GND 2 — VIN– — VOUT– — 4 GND
12 V –> 7.5 V buck (>=5 A)

U3
VBAT 1 — BAT+ — M+ — 6 PUMP_M+
GND 2 — BAT– — M– — 7 PUMP_M–
PUMP_PWM 4 — SIG — SIG_GND 5 — GND
BEC+ 3 (not used)
ESC
Pump ESC (brushed, reversible)
Pump ESC sits dry so any brushed ESC or H–bridge works. BEC wire cut.

PEN3
R2: 20 AWG cables potted (H1, servo, THR sig, wheel sig)

PEN4
R3: 14 AWG cables potted (trunk, thruster pwr, pump motor, wheel pwr)

WET ZONE (flooded hull, open rear endcap, splash areas): Deutsch DT connectors live here

25 A ATC, waterproof holder
F1
VBAT
BT1
3S LiPo (wet, no box)
SW1
GND
Kill switch, low side, >=30 A

Battery side harness: battery + –> F1 –> split to D1 pins 1, 2;
battery – –> SW1 –> split to D1 pins 3, 4.
Splits: solder + adhesive heat–shrink.
Charge through a DT04–4P –> XT60 adapter.

D1A D1B

VBAT	1	1	VBAT
VBAT	2	2	VBAT
GND	3	3	GND
GND	4	4	GND

D1 battery DT06–4S
DT04–4P
D1 battery: 2 contacts per pole = 26 A.
Trunk to rear box: 4 x 14 AWG.

D4A D4B

M4_PWM	1	1	M4_PWM
M5_PWM	2	2	M5_PWM
M6_PWM	3	3	M6_PWM
M7_PWM	4	4	M7_PWM
THR_PWM	5	5	THR_PWM
DRV_L_PWM	6	6	DRV_L_PWM
DRV_R_PWM	7	7	DRV_R_PWM
PUMP_PWM	8	8	PUMP_PWM
LEAK_REAR	9	9	LEAK_REAR
+3V3	10	10	+3V3
GND	11	11	GND
GND	12	12	GND

D4 harness H1 DT06–12S
DT04–12P
D4: main box <–> rear box, 12 x 20 AWG.
Pins 6–7 spare in Plan B.

D3A D3B

M4_PWM	1	1	M4_PWM
SERVO_7V5	2	2	SERVO_7V5
GND	3	3	GND
M5_PWM	4	4	M5_PWM
SERVO_7V5	5	5	SERVO_7V5
GND	6	6	GND
M6_PWM	7	7	M6_PWM
SERVO_7V5	8	8	SERVO_7V5
GND	9	9	GND
M7_PWM	10	10	M7_PWM
SERVO_7V5	11	11	SERVO_7V5
GND	12	12	GND

D3 servos DT06–12S
DT04–12P
D3: 4 servo leads, 12 x 20 AWG.

M4_PWM / SERVO_7V5 / GND — M4 PWM — Rudder servo
M5_PWM / SERVO_7V5 / GND — M5 PWM — Rudder servo
M6_PWM / SERVO_7V5 / GND — M6 PWM — Elevator servo
M7_PWM / SERVO_7V5 / GND — M7 PWM — Elevator servo

D2A D2B

VBAT	1	1	VBAT
GND	2	2	GND
THR_PWM	3	3	THR_PWM
GND	4	4	GND
PUMP_M+	5	5	PUMP_M+
PUMP_M–	6	6	PUMP_M–
	7	7	
	8	8	

D2 aft DT06–8S
DT04–8P
D2: 1–2 thruster ESC 12 V (14 AWG), 3–4 ESC signal (20 AWG).
5–6 pump motor (14 AWG), 7–8 spare (2nd pump).

U2
VBAT 1 — BAT+ — U — 5 — 1
GND 2 — BAT– — V — 6 — 2
THR_PWM 3 — SIG — W — 7 — 3
GND 4 — SIG_GND — BLDC ESC
MT1 BLDC
TD1.2 ESC (waterproof, wet)
TD1.2 thruster
Factory thruster cable stays on the ESC.

M3
PUMP_M+ — 1
PUMP_M– — 2
Ballast pump, 12 V brushed (TBD, <=5 A)

D5A D5B

VBAT	1	1	VBAT
GND	2	2	GND
VBAT	3	3	VBAT
GND	4	4	GND
DRV_L_PWM	5	5	DRV_L_PWM
GND	6	6	GND
DRV_R_PWM	7	7	DRV_R_PWM
GND	8	8	GND

D5 wheels DT06–8S
DT04–8P
D5: 1–4 wheel ESC power (14 AWG).
5–8 wheel ESC signals (20 AWG).

U4
VBAT 1 — BAT+ — M+ — 6 — M1 Driven wheel L (550)
GND 2 — BAT– — M– — 7
DRV_L_PWM 3 — SIG — SIG_GND 5 — GND
BEC+ 4 (not used)
ESC
Wheel ESC L (waterproof)

U5
VBAT 1 — BAT+ — M+ — 6 — M2 Driven wheel R (550)
GND 2 — BAT– — M– — 7
DRV_R_PWM 3 — SIG — SIG_GND 5 — GND
BEC+ 4 (not used)
ESC
Wheel ESC R (waterproof)

Deutsch DT: 13 A per contact, 14–20 AWG, IP68 when every cavity is filled (cavity plugs in unused ones). Source side gets sockets (DT06 plug).

Two wire gauges: HIGH = 14 AWG (12 V, 7.5 V, motor), LOW = 20 AWG (signals, 3.3 V, servo leads).
Only connectors: Deutsch DT in the wet zone + 5.08 mm screw terminals / bus bars inside dryboxes.
Two solder–boards (Board A main drybox, Board B rear drybox). Cables epoxy–potted through drybox

PIE AA–TV

Sheet: /
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Title: AA–TV electrical v7, Plan A: front driven wheels

Size: A2	Date: 2026–11–04	**Rev: v7**
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